

AND if T is sufficient $\Rightarrow I_{\tilde{X}}(\theta) = I_T(\theta)$
 ~~$\Leftarrow$~~

Ex: $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2); \theta = (\mu, \sigma^2)$. Find $I_{\tilde{X}}(\theta)$

$$\ln f(x|\theta) = -\frac{1}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} (x-\mu)^2$$

$$\frac{\partial}{\partial \mu} \ln f = \frac{x-\mu}{\sigma^2} \quad \frac{\partial^2}{\partial \mu^2} \ln f = -\frac{1}{\sigma^2} \quad I_{11}(\theta) = \frac{1}{\sigma^2}$$

$$\frac{\partial}{\partial \sigma^2} \ln f = -\frac{1}{2\sigma^2} + \frac{(x-\mu)^2}{2\sigma^4}$$

$$\frac{\partial^2}{(\partial \sigma^2)^2} \ln f = \frac{1}{2\sigma^4} - \frac{(x-\mu)^2}{\sigma^6}$$

$$I_{22}(\theta) = E\left(-\frac{1}{2\sigma^4} + \frac{(X-\mu)^2}{\sigma^6}\right) = \frac{\sigma^2}{\sigma^6} - \frac{1}{2\sigma^4} = \frac{1}{2\sigma^4}$$

$$\frac{\partial^2}{\partial \mu \partial \sigma^2} \ln f = -\frac{x-\mu}{\sigma^4}$$

$$I_{12}(\theta) = E\left(\frac{x-\mu}{\sigma^4}\right) = 0 = I_{21}(\theta)$$

$$I_{X_1} = \begin{bmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{1}{2\sigma^4} \end{bmatrix}$$

$$\therefore I_{\tilde{X}} = \begin{bmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{bmatrix} > 0$$

HW: $T = (\bar{X}, S^2)$. Compute $I_T(\theta) \underset{\bar{X} \perp S^2}{=} I_{\bar{X}}(\theta) + I_{S^2}(\theta) = I_{\tilde{X}}(\theta)$

$$\begin{aligned} \bar{X} \sim \mathcal{N}(\mu, \frac{\sigma^2}{n}) &\Rightarrow I_{\bar{X}}(\theta) = \begin{bmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{1}{2\sigma^4} \end{bmatrix} \\ \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2 &\Rightarrow I_{S^2}(\theta) = \begin{bmatrix} 0 & 0 \\ 0 & \frac{n-1}{2\sigma^4} \end{bmatrix} \end{aligned} \quad \begin{cases} I_{\tilde{X}}(\theta) - I_{\bar{X}}(\theta) \geq 0 \\ I_{\tilde{X}}(\theta) - I_{S^2}(\theta) \geq 0 \end{cases}$$

Minimal Sufficient Statistics

$\Rightarrow \bar{X}$ not suff for θ
 S^2 not suff for θ

- $X = (X_1, \dots, X_n)$ sample from $f(x|\theta); \theta \in \Theta$
- Any suff. stat. $T(\underline{X})$ leads to data reduction OR induces a sufficient partition of $\mathcal{X}$ such that no info about θ is lost.
 - $T_1(\underline{X}) = (X_1, \dots, X_n)$ gives you the "finest" partition of $\mathcal{X}$
 $T_2(\underline{X}) = (X_{(1)}, \dots, X_{(n)})$ gives you smaller partition of $\mathcal{X}$
 (i.e., # of disjoint classes is smaller)

Ex: $X_1, \dots, X_n$ iid, Bern(θ)

$T_1 = (X_1, \dots, X_n) \Rightarrow$ finest partition of $\mathcal{X}$ into 2^n disjoint classes

$T_2 = (X_1 + X_2, X_3, \dots, X_n) \Rightarrow$ partition of $\mathcal{X}$ into $3 \cdot 2^{n-2}$ disjoint classes -

$T_3 = (X_1 + X_2 + X_3, X_4, \dots, X_n) \Rightarrow$ partition of $\mathcal{X}$ into $4 \cdot 2^{n-3}$ disjoint classes

Clearly T_3 is a fn. of T_2 (as well as T_1)

$T_i = (X_1 + \dots + X_i, X_{i+1}, \dots, X_n) \Rightarrow (i+1) \cdot 2^{n-i}$

T_i is a fn. of $T_j \forall j = 1, \dots, i-1$.

$$T_n = \sum_{i=1}^n X_i \Rightarrow (n+1) \text{ "}$$

It turns out T_n is the "minimal suff." (to be defined)

Q How small can I make a partition such that it still doesn't lose any information about θ ?

Ans. The partition given by "minimal sufficient statistic".

Defⁿ: $T(\underline{X})$ is called a minimal sufficient statistic for θ
[OR for $\mathcal{P} = \{f(x|\theta) : \theta \in \Theta\}$]

(1) $T(\underline{X})$ is sufficient for θ

(2) $T(\underline{X})$ is "minimal", i.e. $T(\underline{X})$ is a function of any other sufficient statistic

[$T(\underline{X}) = h(S(\underline{X}))$ for some fn. h and any suff. stat $S(\underline{X})$]

Ex. (using defⁿ) $X_1, \dots, X_n \text{ iid Bern}(\theta) \quad 0 < \theta < 1$

$T(\underline{X}) = \sum_{i=1}^n X_i$ is min. suff for θ .

Let $\underline{x}, \underline{y} \in \mathcal{X}$

$$\begin{aligned} \text{NFFT} \left(\frac{f(\underline{x}|\theta)}{f(\underline{y}|\theta)} \right) &= \frac{\theta^{\sum x_i} (1-\theta)^{n-\sum x_i}}{\theta^{\sum y_i} (1-\theta)^{n-\sum y_i}} = \left(\frac{\theta}{1-\theta} \right)^{\sum x_i - \sum y_i} \text{ for } \theta \\ &= \frac{h(\underline{x}) g_\theta(S(\underline{x}))}{h(\underline{y}) g_\theta(S(\underline{y}))} \text{ for some suff. stat } S(\underline{X}) \end{aligned}$$

If $\underline{x}, \underline{y} \in \mathcal{X}$ such that $S(\underline{x}) = S(\underline{y})$

$$\frac{f(\underline{x}|\theta)}{f(\underline{y}|\theta)} = \left(\frac{\theta}{1-\theta}\right)^{\sum x_i - \sum y_i} = \frac{h(\underline{x})}{h(\underline{y})} \text{ is free of } \theta$$

$$\Rightarrow \sum_1^n x_i = \sum_1^n y_i$$

$$\Rightarrow T(\underline{x}) = T(\underline{y})$$

We proved: (1) $T(\underline{x}) = \sum_1^n X_i$ is suff. stat for θ (NFFT)

(2) For any sufficient statistic $S(\underline{x})$,
 $S(\underline{x}) = S(\underline{y}) \Rightarrow T(\underline{x}) = T(\underline{y})$

$\Rightarrow T$ is a function of S .

$\Rightarrow T = h(S)$ for some fn. h .

Ex. 2 $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, 1)$. $T(\underline{x}) = \sum X_i$ is a suff stat.
 Is it minimal?

$$\begin{aligned} \text{For } \underline{x}, \underline{y} \in \mathcal{X} \quad \frac{f(\underline{x}|\mu)}{f(\underline{y}|\mu)} &= \frac{e^{-\frac{1}{2} \sum_1^n (x_i - \mu)^2}}{e^{-\frac{1}{2} \sum_1^n (y_i - \mu)^2}} \\ &= e^{\frac{1}{2} (\sum y_i^2 - 2\mu \sum y_i - \sum x_i^2 + 2\mu \sum x_i)} \\ &= e^{\frac{1}{2} (\sum y_i^2 - \sum x_i^2)} e^{\mu (\sum x_i - \sum y_i)} \end{aligned}$$

Consider any suff. stat. S for μ and write

$$\frac{f(\underline{x}|\mu)}{f(\underline{y}|\mu)} = \frac{h(\underline{x}) g_\mu(S(\underline{x}))}{h(\underline{y}) g_\mu(S(\underline{y}))}$$

$$\text{If } S(\underline{x}) = S(\underline{y}) \\ \Rightarrow \frac{f(\underline{x}|\mu)}{f(\underline{y}|\mu)} = e^{\frac{1}{2}(\sum y_i^2 - \sum x_i^2)} e^{\mu(\sum x_i - \sum y_i)} = \frac{h(\underline{x})}{h(\underline{y})} \text{ is free of } \mu$$

$$\Rightarrow \sum x_i = \sum y_i$$

$$\Rightarrow T(\underline{x}) = T(\underline{y})$$

$\therefore T = h(S)$ for some f or h .

Ex 3. $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Gamma}(\alpha, \beta)$ $\theta = (\alpha, \beta) \in \mathbb{R}^+ \times \mathbb{R}^+$

$$f(x|\theta) = \frac{e^{-x/\beta} x^{\alpha-1}}{\beta^\alpha \Gamma(\alpha)} \text{ for } x > 0$$

For $\underline{x}, \underline{y} \in \mathcal{X}$

$$\frac{f(\underline{x}|\theta)}{f(\underline{y}|\theta)} = \frac{e^{-\sum x_i/\beta} \left(\prod_1^n x_i\right)^{\alpha-1}}{e^{-\sum y_i/\beta} \left(\prod_1^n y_i\right)^{\alpha-1}}$$

$$\frac{h(\underline{x}) g(S(\underline{x}))}{h(\underline{y}) g(S(\underline{y}))} = e^{-1/\beta (\sum x_i - \sum y_i)} \left(\frac{\prod_1^n x_i}{\prod_1^n y_i} \right)^{\alpha-1}$$

$\therefore \text{If } S(\underline{x}) = S(\underline{y})$

$\Rightarrow$ free of θ

$$\Rightarrow \sum x_i = \sum y_i, \prod_1^n x_i = \prod_1^n y_i$$

$\Rightarrow (\sum X_i, \prod X_i)$ is min suff.

Result: Any one-one function of minimal sufficient statistic is also minimal sufficient.

Pf: T_1 min suff

$$T_1 = h(S)$$

$$T_2 = \phi(T_1)$$

$\uparrow$
one-one

$$\therefore T_2 = \phi(h(S))$$

Recall (likelihood principle)

If for any x, y , $L(\theta|x) \propto L(\theta|y)$, then inference based on x and y should be same.

Def: Two points x and $y \in \mathcal{X}$ are called "likelihood equivalent" if $L(\theta|x) \propto L(\theta|y)$ (or in other words if " $x \sim y$ ")
 $\frac{f(x|\theta)}{f(y|\theta)} = K(x, y)$ free of θ

" $\sim$ " ($\equiv$ likelihood equivalence) partitions $\mathcal{X}$ into equivalent classes

will prove: Thm the partition induced by likelihood equivalence " $\sim$ " is the minimal sufficient partition

Partition: $A_t = \{x : T(x) = t\}$ where $t \in R(T)$
 $\mathcal{U} = \{A_t : t \in R(T)\}$, a partition of $\mathcal{X}$.

$\downarrow$
range of T